AC9M6M03 • Year 6 Mathematics

Timetables, Itineraries and Journey Duration

Plan connected activities and account for waiting, transfer and overnight time

We are learning to…

Students read 12- and 24-hour schedules, connect services, calculate elapsed time across noon or midnight and test whether an itinerary meets transfer, opening-time and duration constraints.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

interpret and use timetables and itineraries to plan activities and determine the duration of events and journeys

Build a connected journey itinerary

stage	depart	arrive	duration / wait
bus	**08:35**	**09:12**	**37 min**
transfer	09:12	09:28	16 min
train	09:28	11:05	1 h 37 min
total	08:35	11:05	2 h 30 min

Total journey duration includes waiting and transfer time, not only travel stages. Confirm that the connection is achievable.

Plan activities under timetable constraints

identify fixed times → calculate durations → include travel/transition → check overlap → allow buffer → select feasible itinerary → communicate plan

A plan may be mathematically possible but impractical without preparation or delay allowance. State assumptions about time zones and dates where relevant.

Build and annotate the model

Represent timetables, itineraries and journey duration and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Travel times added but waits omitted:** Include every interval from start to finish.
- **Clock digits subtracted as decimals:** Use 60-minute hours and timelines.
- **Arrival before departure ignored:** Check date, midnight and time zone.
- **Minimum transfer assumed safe:** Consider practical buffer when planning.

Quick check

- Find 08:35–11:05.
- Calculate a transfer wait.
- Check overlap.
- Cross midnight.
- Add a practical buffer.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.